

Geometric tests

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Introduction

Geometric predicates are functions that return the sign of a polynomial expression in the input coordinates, encoding geometric relationships such as orientation or sphere containment. It is critical to implement these robustly, as floating-point arithmetic can produce incorrect signs in near-degenerate configurations [1].

To this end, we use the generic floating-point filter framework of [2], which automatically derives tight semi-static error bounds for arbitrary arithmetic expressions at compile time. Each predicate is evaluated first by its floating-point approximation, which is compared against a rigorously derived error bound; if the result is inconclusive, a fallback to exact expansion arithmetic [1] guarantees a correct answer. Underflow protection is enabled throughout.

All of the implemented tests return a value in $\{-1, 0, 1\}$. This can be interpreted as each test answering some geometric question, mapping to {no, uncertain, yes}.

This document first lists and gives a brief interpretation of the implemented tests. The full mathematical derivations and proofs are found in the two appendices.

1 Description of implemented tests

We list here the tests implemented and their interpretation in the case of output 1.

1.1 Primitive tests ($\mathbb{R}^n$)

Note that in the “orient” tests, input order of the first points matters, whereas in the incircle / insphere tests it does not.

Test	Geometric interpretation
<code>test_orientation_2D(a,b,c)</code>	c below oriented line passing by (a, b) (in $\mathbb{R}^2$).
<code>test_orientation_3D(a,b,c,d)</code>	d below oriented plane passing by (a, b, c) (in $\mathbb{R}^3$).
<code>test_incircle(abc[3],d)</code>	d inside circle passing by $\{a, b, c\}$ (in $\mathbb{R}^2$).
<code>test_insphere(abcd[4],e)</code>	e inside sphere passing by $\{a, b, c, d\}$ (in $\mathbb{R}^3$).

With the previous basic tests we can build others, such as `test_point_in_triangle_2D/3D`, `test_point_in_tetrahedron`, and `segment_triangle_intersect_2D/3D`.

These tests have an `EPS_DEG` parameter, controlling the threshold for degenerate cases (i.e. triangle area $< \text{EPS_DEG}$...), and a `TOL` parameter, controlling the threshold for boundary cases (distance from point to boundary of triangle...). If `TOL` = 0, then the tests are exact.

1.2 Orthogonal hypersphere tests ($\mathbb{R}^n \times \mathbb{R}$)

In order to give a proper interpretation of the *power* tests, we first define a weighted point $\hat{x} = (x, w_x) \in (\mathbb{R}^n \times \mathbb{R})$, interpreted as a hypersphere with center x and radius $r_x = \sqrt{w_x}$ (for $w_x > 0$). We define the power distance between two weighted points as

$$\pi(\hat{x}, \hat{y}) = \|x - y\|^2 - w_x - w_y = \|x - y\|^2 - r_x^2 - r_y^2, \quad \hat{x}, \hat{y} \in (\mathbb{R}^n \times \mathbb{R}).$$

The *orthogonal hypersphere* to a set of k linearly independent weighted points $\hat{p}_i$, $1 \leq k \leq n + 1$, is the unique $\hat{S} = (S, w_S)$ satisfying $\pi(\hat{p}_i, \hat{S}) = 0$ for $i = 1, \dots, k$ and with S in the affine hull of the k points. The interpretation of $\pi(\hat{p}, \hat{S}) < 0$ is not straightforward. However, in the case where $w_p = 0$ and $w_S > 0$, $\pi(\hat{p}, \hat{S}) < 0$ implies that p is inside the sphere $\hat{S}$.

In the orthocircle and orthosphere tests, for efficiency, we have implemented each variant $\alpha = 0$ and $\alpha \neq 0$, in 2D the cases $k = 1$, $k = 2$ and in 3D the cases $k = 1$, $k = 2$, $k = 3$.

Test	Geometric interpretation
<code>test_orthosegment(arr[k], $\hat{p}$)</code>	$\pi(\hat{p}, \hat{S}) < 0$, $\hat{S}$ orthosegment of $\{\hat{a}_1, \dots, \hat{a}_k\}$ (in $\mathbb{R} \times \mathbb{R}$).
<code>test_orthocircle(arr[k], $\hat{p}$)</code>	$\pi(\hat{p}, \hat{S}) < 0$, $\hat{S}$ orthocircle of $\{\hat{a}_1, \dots, \hat{a}_k\}$ (in $\mathbb{R}^2 \times \mathbb{R}$).
<code>test_orthosphere(arr[k], $\hat{p}$)</code>	$\pi(\hat{p}, \hat{S}) < 0$, $\hat{S}$ orthosphere of $\{\hat{a}_1, \dots, \hat{a}_k\}$ (in $\mathbb{R}^3 \times \mathbb{R}$).
<code>test_orthosegment_w(arr[k], α)</code>	$w_S < \alpha$, $\hat{S}$ orthosegment of $\{\hat{a}_1, \dots, \hat{a}_k\}$ (in $\mathbb{R} \times \mathbb{R}$).
<code>test_orthocircle_w(arr[k], α)</code>	$w_S < \alpha$, $\hat{S}$ orthocircle of $\{\hat{a}_1, \dots, \hat{a}_k\}$ (in $\mathbb{R}^2 \times \mathbb{R}$).
<code>test_orthosphere_w(arr[k], α)</code>	$w_S < \alpha$, $\hat{S}$ orthosphere of $\{\hat{a}_1, \dots, \hat{a}_k\}$ (in $\mathbb{R}^3 \times \mathbb{R}$).

1.3 Linear Programming tests for Voronoi cells

A *planar constraint* (i, j, k below) is a half-plane in the oblique coordinates of a triangle's plane — an edge of the triangle, or the Voronoi bisector to a neighboring seed; two meet at a boundary point $i \wedge j$. A *spatial constraint* (a, b, c, d below) is a half-space in $\mathbb{R}^3$ — a tetrahedron face, or a Voronoi bisector to a neighboring seed; three meet at a boundary point $a \wedge b \wedge c$.

Test	Geometric interpretation
<code>lp2_D(i,j)</code>	j is a lower bound for the boundary line of i (in $\mathbb{R}^2$).
<code>lp2_feasible(i,j;k)</code>	$i \wedge j$ strictly satisfies k (in $\mathbb{R}^2$).
<code>lp3_D(a,b,c)</code>	a, b, c meet with a fixed right-handed orientation (in $\mathbb{R}^3$).
<code>lp3_feasible(a,b,c;d)</code>	$a \wedge b \wedge c$ strictly satisfies d (in $\mathbb{R}^3$).

Built from these: `lp2_vcell_hits_face` (a boolean, not a sign) tests cell-triangle intersection; `lp3_feasible` drives the tetrahedron clip, and — combined with `lp3_D`, not as any new predicate — also orders reconstructed-surface vertices sharing an edge (§B.4).

A Geometric predicates in $\mathbb{R}^n$ and $\mathbb{R}^n \times \mathbb{R}$

A.1 Orientation

Definition 1 (Generalized cross product). Let $v_1, \dots, v_{n-1} \in \mathbb{R}^n$. Denote by $e_1, \dots, e_n$ the standard basis vectors of $\mathbb{R}^n$ and $v_i^{(j)} \in \mathbb{R}$ the j -th component of v_i . We define $\text{cross}(v_1, \dots, v_{n-1}) \in \mathbb{R}^n$ as the vector resulting from formally expanding the following determinant along the first row:

$$\text{cross}(v_1, \dots, v_{n-1}) = \det \begin{pmatrix} e_1 & \cdots & e_n \\ v_1^{(1)} & \cdots & v_1^{(n)} \\ \vdots & & \vdots \\ v_{n-1}^{(1)} & \cdots & v_{n-1}^{(n)} \end{pmatrix}.$$

In $\mathbb{R}^3$, this reduces to the standard $v_1 \times v_2$.

Proposition 2 (Determinant characterization). *Let $v_1, \dots, v_{n-1} \in \mathbb{R}^n$. Then for any $x \in \mathbb{R}^n$:*

$$\text{cross}(v_1, \dots, v_{n-1}) \cdot x = \det \left(x \mid v_1 \mid \cdots \mid v_{n-1} \right)$$

Proof. Let M_i be the $(n-1) \times (n-1)$ determinant of the matrix with rows $v_1, \dots, v_{n-1}$, with column i deleted. Then by definition, $\text{cross}(v_1, \dots, v_{n-1}) = (M_1, -M_2, \dots, (-1)^{n-1}M_n)$.

Therefore $\text{cross}(v_1, \dots, v_{n-1}) \cdot x = \sum_{i=1}^n (-1)^{1+i} M_i x^{(i)}$, which is precisely the cofactor expansion of $\det(x, v_1, \dots, v_{n-1})$ along the first row, since the cofactor of $x^{(i)}$ in the first row is $(-1)^{1+i} M_i$. $\square$

From this last proposition it's clear that $\text{cross}(v_1, \dots, v_{n-1})$ is perpendicular to each v_i .

Given n affinely independent points, it is easy to see that there exists a unique hyperplane passing through them. The following definition establishes an orientation for it.

Definition 3 (Hyperplane orientation). Let $(p_1, \dots, p_n)$ be an ordered set of n affinely independent points in $\mathbb{R}^n$. The hyperplane H passing through them has normal

$$\mathbf{n} = \text{cross}(p_2 - p_1, p_3 - p_1, \dots, p_n - p_1),$$

so it is precisely the set $H = \{x \in \mathbb{R}^n \mid \mathbf{n} \cdot (x - p_1) = 0\}$. H^+ , H^- are the upper and lower half-spaces, respectively: $H^+ = \{x \in \mathbb{R}^n \mid \mathbf{n} \cdot (x - p_1) > 0\}$, $H^- = \{x \in \mathbb{R}^n \mid \mathbf{n} \cdot (x - p_1) < 0\}$.

Theorem 4 (Orientation predicate). *Let $(p_1, \dots, p_n)$ be an ordered set of n affinely independent points in $\mathbb{R}^n$, and let H be the hyperplane passing through them with normal $\mathbf{n}$. Let $p_{n+1} \in \mathbb{R}^n$ be a query point. Define*

$$\text{orient}_n(p_1, \dots, p_{n+1}) = \det \left(p_1 - p_{n+1} \mid \cdots \mid p_n - p_{n+1} \right).$$

Then $\text{orient}_n(p_1, \dots, p_{n+1}) = -\mathbf{n} \cdot (p_{n+1} - p_1)$ and in particular:

- $\text{orient}_n < 0 \iff p_{n+1} \in H^+$ (it lies on the **upper side** of H).
- $\text{orient}_n = 0 \iff p_{n+1} \in H$ (**affinely dependent**).
- $\text{orient}_n > 0 \iff p_{n+1} \in H^-$ (it lies on the **lower side** of H).

Proof. By proposition 2, we have $\mathbf{n} \cdot (p_{n+1} - p_1) = \det \begin{pmatrix} p_{n+1} - p_1 & p_2 - p_1 & \cdots & p_n - p_1 \end{pmatrix}$. Subtracting the first row to all other rows, we obtain $\det \begin{pmatrix} p_{n+1} - p_1 & p_2 - p_{n+1} & \cdots & p_n - p_{n+1} \end{pmatrix}$. Then changing the sign of the first row, $-\det \begin{pmatrix} p_1 - p_{n+1} & p_2 - p_{n+1} & \cdots & p_n - p_{n+1} \end{pmatrix}$. Thus, $\mathbf{n} \cdot (p_{n+1} - p_1) = -\text{orient}_n(p_1, \dots, p_{n+1})$ and its sign directly encodes the belonging of p_{n+1} to H , H^+ or H^- . $\square$

This convention coincides with that of Shewchuk [1], where $\text{orient}_n > 0$ corresponds to p_{n+1} lying on the negative side of H . When $p_1, \dots, p_n$ are affinely dependent, $\text{orient}_n = 0$ regardless of p_{n+1} , so the predicate is degenerate and carries no geometric information about the position of p_{n+1} . Finally note that by performing row operations, we can rewrite as $\text{orient}_n(p_1, \dots, p_{n+1}) = (-1)^n \det \begin{pmatrix} p_2 - p_1 & \cdots & p_{n+1} - p_1 \end{pmatrix}$.

A.2 Orthogonal hypersphere power

Definition 5 (Weighted point). A weighted point in $\mathbb{R}^n$ is a pair $(p, w) \in \mathbb{R}^n \times \mathbb{R}$, where p is called the **center** and $w \in \mathbb{R}$ the **weight**.

A weighted point (p, w) can be interpreted as a hypersphere with center p and radius $r = \sqrt{w}$ when $w > 0$. For $w < 0$, this interpretation leads to the existence of imaginary radii. The case $w = 0$ simply corresponds to a point.

Definition 6 (Power distance). The power distance between two weighted points $\hat{p} = (p, w_p)$ and $\hat{q} = (q, w_q)$ in $\mathbb{R}^n$ is defined as:

$$\pi(\hat{p}, \hat{q}) = \|p - q\|^2 - w_p - w_q.$$

The power distance is symmetric in $\hat{p}$ and $\hat{q}$. When $w_p = 0$, it reduces to the classical power of a point p with respect to the sphere $\hat{q} = (q, w_q)$: $\pi(p, \hat{q}) = \|p - q\|^2 - w_q$, which is negative if p lies inside the sphere, zero if p lies on it, and positive if p lies outside.

Definition 7 (Orthogonal hypersphere). Let $\hat{p}_1, \dots, \hat{p}_k$ be k weighted points in $\mathbb{R}^n$ with $1 \leq k \leq n + 1$, whose centers are affinely independent. The orthogonal hypersphere $\hat{S}$ of these weighted points is the unique weighted point $\hat{S} = (p_S, w_S)$ such that:

$$\begin{cases} \pi(\hat{p}_i, \hat{S}) = 0 & \forall i = 1, \dots, k \\ \text{its center } p_S \text{ lies in the affine hull of } p_1, \dots, p_k. \end{cases}$$

When $k = n + 1$, the affine hull of $p_1, \dots, p_{n+1}$ is all of $\mathbb{R}^n$, so the center constraint is vacuous and the orthogonal hypersphere is determined solely by the $n + 1$ orthogonality conditions. When all weights are zero, $\hat{S}$ reduces to the **circumscribed hypersphere**.

Proposition 8 (Existence and uniqueness). *Let $\hat{p}_1, \dots, \hat{p}_k$ be k weighted points in $\mathbb{R}^n$ with $1 \leq k \leq n + 1$, whose centers are affinely independent. Their orthogonal hypersphere exists and is unique.*

Proof. Subtracting the $i = 1$ orthogonality condition $\pi(\hat{p}_i, \hat{S}) = 0$ from the remaining $k - 1$ eliminates w_S , yielding $k - 1$ linear equations in p_S : $2(p_i - p_1) \cdot p_S = \|p_i\|^2 - \|p_1\|^2 - (w_i - w_1)$, $i = 2, \dots, k$. Restricting p_S to the affine hull of $p_1, \dots, p_k$ reduces the system to a $(k - 1) \times (k - 1)$ linear system whose coefficient matrix is the Gram matrix of $p_2 - p_1, \dots, p_k - p_1$, which is positive definite by affine independence. This uniquely determines p_S , and then $w_S = \|p_1 - p_S\|^2 - w_1$ is uniquely determined. $\square$

Definition 9 (Paraboloid lifting map). The **paraboloid lifting map** $\phi : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^{n+1}$ is defined by:

$$\phi(\hat{p}) = \phi(p, w) = (p, \|p\|^2 - w) \in \mathbb{R}^{n+1}.$$

For an unweighted point, $\phi(\hat{p})$ lies on the paraboloid $\mathcal{K} = \{(x, z) \in \mathbb{R}^{n+1} \mid z = \|x\|^2\}$. A nonzero weight w shifts the last coordinate vertically by $-w$, moving the lifted point below the paraboloid for $w > 0$ and above it for $w < 0$.

Lemma 10 (Lift and power distance). *Let $\hat{q} = (q, w_q)$ be a weighted point in $\mathbb{R}^n$. Define $\Pi \subset \mathbb{R}^{n+1}$ to be the hyperplane with normal $\mathbf{n}_\Pi = (-2q, 1)$ passing by $p_\Pi = (q, \|q\|^2 + w_q)$:*

$$\Pi = \{r \in \mathbb{R}^{n+1} \mid \mathbf{n}_\Pi \cdot (r - p_\Pi) = 0\}, \quad \text{with corresponding halfspaces } \Pi^+, \Pi^-.$$

Then, for any weighted point $\hat{x} \in \mathbb{R}^n \times \mathbb{R}$, the paraboloid lifting map satisfies

$$\phi(\hat{x}) \in \Pi, \Pi^-, \Pi^+ \iff \pi(\hat{x}, \hat{q}) = 0, < 0, > 0 \quad \text{correspondingly.}$$

Proof. Note that developing, we can write Π as $\Pi = \{(x, z) \in \mathbb{R}^{n+1} \mid z = 2q \cdot x - \|q\|^2 + w_q\}$. For any weighted point $\hat{x} = (x, w_x)$, compute the signed vertical distance of $\phi(\hat{x}) = (x, \|x\|^2 - w_x)$ from Π :

$$\delta_\Pi(\hat{x}) = (\|x\|^2 - w_x) - (2q \cdot x - \|q\|^2 + w_q)$$

Expanding and rearranging: $\delta_\Pi(\hat{x}) = \|x\|^2 - 2q \cdot x + \|q\|^2 - w_x - w_q = \|x - q\|^2 - w_x - w_q = \pi(\hat{x}, \hat{q})$. Therefore $\delta_\Pi(\hat{x}) = \pi(\hat{x}, \hat{q})$ for all $\hat{x} \in \mathbb{R}^n \times \mathbb{R}$. $\square$

Actually there is a one-to-one correspondence between weighted points orthogonal to $\hat{q}$ and points in Π , weighted points with negative power distance and points Π^+ , etc. Since $\mathbf{n}$ has positive last component, Π^+ is always “above” in the z -direction.

Theorem 11 (Orthogonal hypersphere power predicate). *Let $\hat{p}_1, \dots, \hat{p}_k$ be k weighted points in $\mathbb{R}^n$ with $2 \leq k \leq n + 1$, whose centers are affinely independent, and consider their orthogonal hypersphere $\hat{S}$. Let $\hat{p}_{k+1}$ be a query weighted point. Let $e_{j_1}, \dots, e_{j_{n+1-k}}$ be standard basis vectors of $\mathbb{R}^n$ s.t.:*

$$\mathbf{v}_s = \text{CROSS}(p_2 - p_1, \dots, p_k - p_1, e_{j_1}, \dots, \widehat{e_{j_s}}, \dots, e_{j_{n+1-k}})$$

are linearly independent vectors in $\mathbb{R}^n$, where $\widehat{e_{j_s}}$ denotes omission of e_{j_s} . Define:

$$\begin{cases} \ell_i = \|p_i - p_{k+1}\|^2 - (w_i - w_{k+1}), & i = 1, \dots, k \\ \mu_s = 2 \det \left(p_1 - p_{k+1} \mid \dots \mid p_k - p_{k+1} \mid e_{j_1} \mid \dots \mid \widehat{e_{j_s}} \mid \dots \mid e_{j_{n+1-k}} \right), & s = 1, \dots, n + 1 - k \end{cases}$$

and the $(n + 1) \times (n + 1)$ determinant:

$$D = \det \begin{pmatrix} p_1 - p_{k+1} & \ell_1 \\ \vdots & \vdots \\ p_k - p_{k+1} & \ell_k \\ \mathbf{v}_1 & \mu_1 \\ \vdots & \vdots \\ \mathbf{v}_{n+1-k} & \mu_{n+1-k} \end{pmatrix}.$$

Define also the $n \times n$ matrix collecting the direction vectors of the construction:

$$A = \left(p_2 - p_1 \mid \cdots \mid p_k - p_1 \mid e_{j_1} \mid \cdots \mid e_{j_{n+1-k}} \right).$$

Then, we have:

$$\text{sign}(\pi(\hat{p}_{k+1}, \hat{S})) = \begin{cases} (-1)^{\frac{n-k}{2}} \cdot \text{sign}(D), & n - k \text{ even} \\ (-1)^{n+\frac{n-k-1}{2}} \cdot \text{sign}(\det A) \cdot \text{sign}(D), & n - k \text{ odd} \end{cases}$$

Remark 12 (Orientation form of the factor, and why the anchor matters). $\det A$ is the orientation of the direction basis completing the affine hull of the centers, and it is manifestly translation-invariant – as it must be, since $\pi(\hat{p}_{k+1}, \hat{S})$ and D are (every entry of D is a coordinate difference or a direction). It can equivalently be written as an orientation test, *provided* the artificial directions are anchored at a point of the configuration, like the points $p_1 + \mathbf{v}_s$ in the proof: by the row-operation identity of the orientation predicate,

$$\text{orient}_n(p_1, \dots, p_k, p_1 + e_{j_1}, \dots, p_1 + e_{j_{n+1-k}}) = (-1)^n \det A.$$

The anchor is essential: the unanchored expression $\text{orient}_n(p_1, \dots, p_k, e_{j_1}, \dots, e_{j_{n+1-k}})$ places the artificial points at the *origin* and is therefore position-dependent (e.g. for $n = 3, k = 2$, translating a and b changes its sign but not that of $\det A = (b - a) \cdot (e_{j_1} \times e_{j_2})$), so it cannot appear in the formula; note the two coincide exactly when p_1 is at the origin, which makes the mistake easy to miss. For $k = n + 1$ no artificial vectors remain and $\det A = (-1)^n \text{orient}_n(p_1, \dots, p_{n+1})$, recovering the familiar orientation form. In the implemented case $n = 3, k = 2$, $\text{sign}(\det A)$ is the sign of a single coordinate difference of $b - a$ (see the case expansions below), computed exactly by one comparison.

Proof.

Step 1: Lifted input points lie on Π . By lemma 10, since $\pi(\hat{p}_i, \hat{S}) = 0$ for $i = 1, \dots, k$, the lifted points $\phi(\hat{p}_1), \dots, \phi(\hat{p}_k)$ all lie on the hyperplane $\Pi \subset \mathbb{R}^{n+1}$ associated to $\hat{S}$.

Step 2: Lifted artificial points lie on Π . Define the artificial weighted points $\hat{q}_s = (p_1 + \mathbf{v}_s, w_1 + \|\mathbf{v}_s\|^2)$ for $s = 1, \dots, n + 1 - k$. We claim they are orthogonal to $\hat{S}$, $\pi(\hat{q}_s, \hat{S}) = 0$. Expanding:

$$\pi(\hat{q}_s, \hat{S}) = \|p_1 + \mathbf{v}_s - p_S\|^2 - w_1 - \|\mathbf{v}_s\|^2 - w_S = \underbrace{\|p_1 - p_S\|^2 - w_1 - w_S}_{=\pi(\hat{p}_1, \hat{S})=0} + 2 \underbrace{(p_1 - p_S) \cdot \mathbf{v}_s}_{=0}.$$

The first term vanishes by definition of $\hat{S}$. The second term vanishes because p_S lies in the affine hull of $p_1, \dots, p_k$, so $p_1 - p_S = \sum_{i=2}^k \lambda_i (p_i - p_1)$, and by the cross product's orthogonality property, $(p_i - p_1) \cdot \mathbf{v}_s = 0$ for $i = 2, \dots, k$ since each $p_i - p_1$ is an input of $\mathbf{v}_s$.

Step 3: D equals orient_{n+1} of the lifted points. Consider the $(n + 1) \times (n + 1)$ determinant (written in transposed manner here for compactness, but think of it as constructed row by row):

$$\begin{aligned} \Delta &= \text{orient}_{n+1}(\phi(\hat{p}_1), \dots, \phi(\hat{p}_k), \phi(\hat{q}_1), \dots, \phi(\hat{q}_{n+1-k}), \phi(\hat{p}_{k+1})) \\ &= \det \left(\phi(\hat{p}_1) - \phi(\hat{p}_{k+1}) \mid \dots \mid \phi(\hat{p}_k) - \phi(\hat{p}_{k+1}) \mid \phi(\hat{q}_1) - \phi(\hat{p}_{k+1}) \mid \dots \mid \phi(\hat{q}_{n+1-k}) - \phi(\hat{p}_{k+1}) \right). \end{aligned}$$

The input rows have the form $(p_i - p_{k+1}, \|p_i\|^2 - w_i - \|p_{k+1}\|^2 + w_{k+1})$, which can be converted to $(p_i - p_{k+1}, \ell_i)$ after column operations subtracting $2p_{k+1}^{(j)}$ times column j from the last column. The

artificial rows have the form $(p_1 - p_{k+1} + \mathbf{v}_s, \|p_1\|^2 + 2p_1 \cdot \mathbf{v}_s - w_1 - \|p_{k+1}\|^2 + w_{k+1})$. After the column operations, the last entry of the artificial row simplifies to

$$\|p_1\|^2 + 2p_1 \cdot \mathbf{v}_s - w_1 - \|p_{k+1}\|^2 + w_{k+1} - 2p_{k+1} \cdot (p_1 - p_{k+1} + \mathbf{v}_s) = \ell_1 + 2(p_1 - p_{k+1}) \cdot \mathbf{v}_s = \ell_1 + \mu_s.$$

The last equality is easy to see by applying 2 and trivial row manipulations. Therefore the artificial rows become $(p_1 - p_{k+1} + \mathbf{v}_s, \ell_1 + \mu_s)$. Subtracting the first input row $(p_1 - p_{k+1}, \ell_1)$ from each artificial row gives rows of the form $(\mathbf{v}_s, \mu_s)$. Hence $\Delta = D$.

Step 4: Sign relationship. By the orient theorem in $\mathbb{R}^{n+1}$, the sign of Δ tells us on which side of the hyperplane through $(\phi(\hat{p}_1), \dots, \phi(\hat{p}_k), \phi(\hat{q}_1), \dots, \phi(\hat{q}_{n+1-k}))$ the point $\phi(\hat{p}_{k+1})$ lies, relative to the orientation *induced by the ordering* of those points. To compare this to Π as defined by $\mathbf{n}_\Pi = (-2p_S, 1)$ in lemma 10, we can project the induced normal to e_{n+1} (since $\mathbf{n}_\Pi$ points upwards in the last component):

$$\begin{aligned} (\text{induced normal}) \cdot e_{n+1} &= \text{cross}(\phi(\hat{p}_2) - \phi(\hat{p}_1), \dots, \phi(\hat{q}_{n+1-k}) - \phi(\hat{p}_1)) \cdot (0, \dots, 0, 1) \\ &= \det \left(\begin{array}{c|c} (0, \dots, 0, 1) & \phi(\hat{p}_2) - \phi(\hat{p}_1) \mid \dots \mid \phi(\hat{q}_{n+1-k}) - \phi(\hat{p}_1) \end{array} \right) \\ &= (-1)^n \det \left(\begin{array}{c|c} p_2 - p_1 & \dots \mid q_{n+1-k} - p_1 \end{array} \right) \\ &= (-1)^n (-1)^{n-1} \det \left(\begin{array}{c|c} p_1 - q_{n+1-k} & \dots \mid q_{n-k} - q_{n+1-k} \end{array} \right) \\ &= -\text{orient}_n(p_1, \dots, p_k, q_1, \dots, q_{n+1-k}) \end{aligned}$$

Thus, the induced orientation will point in the same direction as $\mathbf{n}_\Pi$ (and the oriented plane will be the same as Π) iff $-\text{orient}_n(p_1, \dots, q_{n+1-k}) > 0$. Combining with lemma 10, we have:

$$\text{sign}(\pi(\hat{p}_{k+1}, \hat{S})) = -\text{sign}(\text{orient}_n(p_1, \dots, p_k, p_1 + \mathbf{v}_1, \dots, p_1 + \mathbf{v}_{n+1-k})) \cdot \text{sign}(D).$$

Step 5: Rewrite $\text{orient}_n(\dots)$: Define $\delta_i = (p_{i+1} - p_1)$ and consider the $n \times n$ matrices

$$A = \begin{pmatrix} \delta_1 \\ \vdots \\ \delta_{k-1} \\ e_{j_1} \\ \vdots \\ e_{j_{n+1-k}} \end{pmatrix}, \quad B = \begin{pmatrix} \delta_1 \\ \vdots \\ \delta_{k-1} \\ \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_{n+1-k} \end{pmatrix}.$$

(A is the transpose of the matrix in the theorem statement, which leaves the determinant unchanged.) Note that $\det(B) = (-1)^n \text{orient}_n(p_1, \dots, p_k, p_1 + \mathbf{v}_1, \dots)$ by the row-operation identity of the orientation predicate (all its arguments are genuine points). Combining with step 4,

$$\text{sign}(\pi(\hat{p}_{k+1}, \hat{S})) = (-1)^{n+1} \text{sign}(\det B) \cdot \text{sign}(D),$$

and the remainder of the proof works directly with determinants. Consider the product AB^T . By the identity $w \cdot \text{cross}(u_1, \dots, u_{n-1}) = \det(w \mid u_1 \mid \dots \mid u_{n-1})$:

- For $1 \leq i, j \leq k-1$: $(AB^T)_{ij} = \delta_i \cdot \delta_j = G_{ij}$, the Gram matrix of $\delta_1, \dots, \delta_{k-1}$.
- For $1 \leq i \leq k-1$ and $k \leq j \leq n$: $(AB^T)_{ij} = \delta_i \cdot \mathbf{v}_{j-k+1} = 0$.
- For $k \leq i \leq n$ and $1 \leq j \leq k-1$: $(AB^T)_{ij} = e_{j_{i-k+1}} \cdot \delta_j = 0$.
- For $k \leq i, j \leq n$: $(AB^T)_{ij} = e_{j_{i-k+1}} \cdot \mathbf{v}_{j-k+1}$. By the cross product identity this equals

$$\det \left(\begin{array}{c|c} e_{j_{i-k+1}} & \delta_1 \mid \dots \mid \delta_{k-1} \mid e_{j_1} \mid \dots \mid \widehat{e_{j_{j-k+1}}} \mid \dots \mid e_{j_{n+1-k}} \end{array} \right).$$

If $i \neq j$, then $e_{j_{i-k+1}}$ already appears and hence $(AB^T)_{ij} = 0$. If $i = j$, the row e_{j_s} is moved to its position i in A , requiring $i-1$ adjacent row swaps. Therefore $(AB^T)_{ii} = (-1)^{i-1} \det(A)$.

Therefore AB^T is block diagonal with blocks G and $\text{diag}((-1)^{k-1}, (-1)^k, \dots, (-1)^{n-1}) \det(A)$. Taking determinants of both sides and using $\sum_{i=k}^n (i-1) = \frac{(n+1-k)(n+k-2)}{2}$:

$$\det(A) \cdot \det(B) = \det(G) \cdot (-1)^{\frac{(n+1-k)(n+k-2)}{2}} \det(A)^{n+1-k}.$$

Dividing by $\det(A)$ (nonzero by linear independence of δ_i and e_{j_s}):

$$\det(B) = (-1)^{\frac{(n+1-k)(n+k-2)}{2}} \det(G) \cdot \det(A)^{n-k}.$$

Since $\det(G) > 0$, taking signs:

$$\text{sign}(\det B) = (-1)^{\frac{(n+1-k)(n+k-2)}{2}} \cdot \text{sign}(\det(A)^{n-k}).$$

Substituting into the relation obtained at the beginning of this step,

$$\text{sign}(\pi(\hat{p}_{k+1}, \hat{S})) = (-1)^E \text{sign}(\det(A)^{n-k}) \cdot \text{sign}(D), \quad E = \frac{(n+1-k)(n+k-2)}{2} + n + 1.$$

Finally, observe that $\text{sign}(x^N)$ simplifies to $+1$ if N is even, and to $\text{sign}(x)$ if N is odd.

Step 6: simplify $(-1)^E$: Define $m = n - k$. Substituting and noting $n + k - 2 = 2n - m - 2$, the exponent becomes

$$E = \frac{(m+1)(2n-m-2)}{2} + n + 1.$$

- **Case m even.** Write $m = 2j$. Then $(m+1)$ is odd and $(2n-m-2)$ is even, so the fraction is an integer. Reducing modulo 2:

$$\frac{(2j+1)(2n-2j-2)}{2} \equiv (2j+1)(n-j-1) \equiv n-j-1 \pmod{2}.$$

Hence

$$E \equiv (n-j-1) + (n+1) = 2n-j \equiv j \pmod{2},$$

so $(-1)^E = (-1)^j = (-1)^{(n-k)/2}$; the $\det A$ factor is absent since $\text{sign}(\det A)^{n-k} = +1$ for $n-k$ even.

- **Case m odd.** Write $m = 2j + 1$. Then $m + 1 = 2(j + 1)$, so

$$\frac{(m+1)(2n-m-2)}{2} = (j+1)(2n-2j-3) \equiv j+1 \pmod{2}.$$

Hence

$$E \equiv (j+1) + (n+1) = n+j+2 \equiv n+j \pmod{2},$$

so $(-1)^E = (-1)^{n+j} = (-1)^{n+(n-k-1)/2}$.

□

Case $k=n+1$

In this case, $n - k = -1$ is odd and no artificial basis vectors remain: A has columns $p_2 - p_1, \dots, p_{n+1} - p_1$, so $\det A = (-1)^n \text{orient}_n(p_1, \dots, p_{n+1})$, and the prefactor is $(-1)^{n+\frac{n-k-1}{2}} = (-1)^{n-1}$. Combining the two, $(-1)^{n-1} \cdot (-1)^n = -1$, and the theorem simplifies to

$$D = \det \begin{pmatrix} p_1 - p_{n+2} & \|p_1 - p_{n+2}\|^2 - (w_1 - w_{n+2}) \\ \vdots & \vdots \\ p_{n+1} - p_{n+2} & \|p_{n+1} - p_{n+2}\|^2 - (w_{n+1} - w_{n+2}) \end{pmatrix}$$

$$\text{sign}(\pi(\hat{p}_{n+2}, \hat{S})) = -\text{sign}(\text{orient}_n(p_1, \dots, p_{n+1})) \cdot \text{sign}(D)$$

Furthermore, in the case where all weights are 0, D simplifies to the regular “insphere” predicate described by [1]. If $D > 0$ and $\text{orient}_n(p_1, \dots, p_{n+1}) > 0$, then $\pi(p_{n+2}, \hat{S}) < 0$ and thus p_{n+2} is inside the orthosphere of $(\hat{p}_1, \dots, \hat{p}_{n+1})$.

Case n=2, k=2

Consider $\hat{a} = (a, w_a), \hat{b} = (b, w_b) \in \mathbb{R}^2 \times \mathbb{R}$, and the query point $\hat{c} = (c, w_c) \in \mathbb{R}^2 \times \mathbb{R}$. Then, $\mathbf{v}_1 = \text{cross}(b - a) = (b_y - a_y, a_x - b_x)$, and $\mu_1 = 2(a_x - c_x)(b_y - c_y) - 2(a_y - c_y)(b_x - c_x)$. Also note that $n - k = 0$ is even, and $(-1)^{\frac{n-k}{2}} = (-1)^0 = +1$, so

$$D = \det \begin{pmatrix} a_x - c_x & a_y - c_y & (a_x - c_x)^2 + (a_y - c_y)^2 - (w_a - w_c) \\ b_x - c_x & b_y - c_y & (b_x - c_x)^2 + (b_y - c_y)^2 - (w_b - w_c) \\ b_y - a_y & a_x - b_x & 2(a_x - c_x)(b_y - c_y) - 2(a_y - c_y)(b_x - c_x) \end{pmatrix},$$

$$\text{sign}(\pi(\hat{c}, \hat{S})) = \text{sign}(D).$$

Case n=3, k=3

Consider $\hat{a}, \hat{b}, \hat{c} \in \mathbb{R}^3 \times \mathbb{R}$, and the query point $\hat{d} \in \mathbb{R}^3 \times \mathbb{R}$. Then, $\mathbf{v}_1 = \text{cross}(b - a, c - a)$. The components of this usual vector product in $\mathbb{R}^3$ are $\mathbf{v}_1^x = (b_y - a_y)(c_z - a_z) - (b_z - a_z)(c_y - a_y)$, $\mathbf{v}_1^y = (b_z - a_z)(c_x - a_x) - (b_x - a_x)(c_z - a_z)$, and $\mathbf{v}_1^z = (b_x - a_x)(c_y - a_y) - (b_y - a_y)(c_x - a_x)$. $\mu_1 = 2 \det (a - d \mid b - d \mid c - d)$. Also note that $n - k = 0$ is even, and $(-1)^{\frac{n-k}{2}} = +1$, so

$$D = \det \begin{pmatrix} a_x - d_x & a_y - d_y & a_z - d_z & (a_x - d_x)^2 + (a_y - d_y)^2 + (a_z - d_z)^2 - (w_a - w_d) \\ b_x - d_x & b_y - d_y & b_z - d_z & (b_x - d_x)^2 + (b_y - d_y)^2 + (b_z - d_z)^2 - (w_b - w_d) \\ c_x - d_x & c_y - d_y & c_z - d_z & (c_x - d_x)^2 + (c_y - d_y)^2 + (c_z - d_z)^2 - (w_c - w_d) \\ \mathbf{v}_1^x & \mathbf{v}_1^y & \mathbf{v}_1^z & 2 \det (a - d \mid b - d \mid c - d) \end{pmatrix},$$

$$\text{sign}(\pi(\hat{d}, \hat{S})) = \text{sign}(D).$$

Case n=3, k=2

Consider $\hat{a}, \hat{b} \in \mathbb{R}^3 \times \mathbb{R}$, and the query point $\hat{c} \in \mathbb{R}^3 \times \mathbb{R}$. We have to choose e_{j_1}, e_{j_2} , basis vectors of $\mathbb{R}^3$, least parallel to $b - a$. $\mathbf{v}_1 = \text{cross}(b - a, e_{j_2})$, $\mathbf{v}_2 = \text{cross}(b - a, e_{j_1})$. Define $\delta a = a - c$, $\delta b = b - c$. Then $\mu_1 = 2 \det (\delta a \mid \delta b \mid e_{j_2})$, $\mu_2 = 2 \det (\delta a \mid \delta b \mid e_{j_1})$. Also note that $n - k = 1$ is odd, the prefactor is $(-1)^{n+\frac{n-k-1}{2}} = (-1)^{3+0} = -1$, and $\det A = \det (b - a \mid e_{j_1} \mid e_{j_2}) = (b - a) \cdot (e_{j_1} \times e_{j_2})$, i.e. the component of $b - a$ left out of the basis pair. So

$$D = \det \begin{pmatrix} \delta a_x & \delta a_y & \delta a_z & \delta a_x^2 + \delta a_y^2 + \delta a_z^2 - (w_a - w_c) \\ \delta b_x & \delta b_y & \delta b_z & \delta b_x^2 + \delta b_y^2 + \delta b_z^2 - (w_b - w_c) \\ \mathbf{v}_1^x & \mathbf{v}_1^y & \mathbf{v}_1^z & \mu_1 \\ \mathbf{v}_2^x & \mathbf{v}_2^y & \mathbf{v}_2^z & \mu_2 \end{pmatrix},$$

$$\text{sign}(\pi(\hat{c}, \hat{S})) = -\text{sign}((b - a) \cdot (e_{j_1} \times e_{j_2})) \cdot \text{sign}(D).$$

Note that the orientation factor $-\det A$ is a single coordinate difference of $b - a$ (see the per-case expressions below), which is computed exactly by one floating-point comparison – unlike D , it needs no robust evaluation. Since the basis pair is chosen least parallel to $b - a$, this is the component of $b - a$ largest in magnitude, so the factor vanishes only when $a = b$.

- $\mathbf{e}_{j_1} = \mathbf{e}_1, \mathbf{e}_{j_2} = \mathbf{e}_2$:

- $\mathbf{v}_1 = (a_z - b_z, 0, b_x - a_x)$, $\mathbf{v}_2 = (0, b_z - a_z, a_y - b_y)$.
- $\mu_1 = 2\delta a_z \delta b_x - 2\delta a_x \delta b_z$, $\mu_2 = 2\delta a_y \delta b_z - 2\delta a_z \delta b_y$.

$$D = \begin{vmatrix} \delta a_x & \delta a_y \\ \delta b_x & \delta b_y \end{vmatrix} \begin{vmatrix} \mathbf{v}_1^z & \mu_1 \\ \mathbf{v}_2^z & \mu_2 \end{vmatrix} + \mathbf{v}_2^y \left(\delta a_x \begin{vmatrix} \delta b_z & \ell_b \\ \mathbf{v}_1^z & \mu_1 \end{vmatrix} - \delta b_x \begin{vmatrix} \delta a_z & \ell_a \\ \mathbf{v}_1^z & \mu_1 \end{vmatrix} \right) + \mathbf{v}_1^x \begin{vmatrix} \delta a_y & \delta a_z & \ell_a \\ \delta b_y & \delta b_z & \ell_b \\ \mathbf{v}_2^y & \mathbf{v}_2^z & \mu_2 \end{vmatrix}$$

$$- \det A = -(b - a) \cdot (e_1 \times e_2) = a_z - b_z$$

• $\mathbf{e}_{j_1} = \mathbf{e}_2, \mathbf{e}_{j_2} = \mathbf{e}_3$:

- $\mathbf{v}_1 = (b_y - a_y, a_x - b_x, 0), \mathbf{v}_2 = (a_z - b_z, 0, b_x - a_x).$
- $\mu_1 = 2\delta a_x \delta b_y - 2\delta a_y \delta b_x, \mu_2 = 2\delta a_z \delta b_x - 2\delta a_x \delta b_z.$

$$D = \begin{vmatrix} \delta a_y & \delta a_z \\ \delta b_y & \delta b_z \end{vmatrix} \begin{vmatrix} \mathbf{v}_1^x & \mu_1 \\ \mathbf{v}_2^x & \mu_2 \end{vmatrix} + \mathbf{v}_2^z \left(\delta a_y \begin{vmatrix} \delta b_x & \ell_b \\ \mathbf{v}_1^x & \mu_1 \end{vmatrix} - \delta b_y \begin{vmatrix} \delta a_x & \ell_a \\ \mathbf{v}_1^x & \mu_1 \end{vmatrix} \right) + \mathbf{v}_1^y \begin{vmatrix} \delta a_z & \delta a_x & \ell_a \\ \delta b_z & \delta b_x & \ell_b \\ \mathbf{v}_2^z & \mathbf{v}_2^x & \mu_2 \end{vmatrix}$$

$$- \det A = -(b - a) \cdot (e_2 \times e_3) = a_x - b_x$$

• $\mathbf{e}_{j_1} = \mathbf{e}_3, \mathbf{e}_{j_2} = \mathbf{e}_1$:

- $\mathbf{v}_1 = (0, b_z - a_z, a_y - b_y), \mathbf{v}_2 = (b_y - a_y, a_x - b_x, 0).$
- $\mu_1 = 2\delta a_y \delta b_z - 2\delta a_z \delta b_y, \mu_2 = 2\delta a_x \delta b_y - 2\delta a_y \delta b_x.$

$$D = \begin{vmatrix} \delta a_z & \delta a_x \\ \delta b_z & \delta b_x \end{vmatrix} \begin{vmatrix} \mathbf{v}_1^y & \mu_1 \\ \mathbf{v}_2^y & \mu_2 \end{vmatrix} + \mathbf{v}_2^x \left(\delta a_z \begin{vmatrix} \delta b_y & \ell_b \\ \mathbf{v}_1^y & \mu_1 \end{vmatrix} - \delta b_z \begin{vmatrix} \delta a_y & \ell_a \\ \mathbf{v}_1^y & \mu_1 \end{vmatrix} \right) + \mathbf{v}_1^z \begin{vmatrix} \delta a_x & \delta a_y & \ell_a \\ \delta b_x & \delta b_y & \ell_b \\ \mathbf{v}_2^x & \mathbf{v}_2^y & \mu_2 \end{vmatrix}$$

$$- \det A = -(b - a) \cdot (e_3 \times e_1) = a_y - b_y$$

When $p_1, \dots, p_k$ are affinely dependent, the orthosphere $\hat{S}$ is not uniquely defined and $D = 0$ (since the first k rows of D become linearly dependent), so $\pi(\hat{p}_{k+1}, \hat{S}) = 0$ but this case carries no geometrical meaning.

An alternative: clearing the Gram denominator

The theorem above computes $\text{sign}(\pi)$ through the lifted determinant D . A second route solves the orthocenter system directly by Cramer's rule and clears its denominator, which is a *positive* quantity – so no orientation factor of any kind appears.

Proposition 13 (Cleared form of the power predicate). *Let $\hat{p}_1, \dots, \hat{p}_k$ be k weighted points in $\mathbb{R}^n$ with $2 \leq k \leq n + 1$, whose centers are affinely independent, let $\hat{S}$ be their orthogonal hypersphere and $\hat{q} = (q, w_q)$ a query weighted point. Define $\delta_i = p_{i+1} - p_1, t = q - p_1$, the Gram matrix $G_{ij} = \delta_i \cdot \delta_j$, and the vectors $u, b \in \mathbb{R}^{k-1}$ with $u_i = t \cdot \delta_i$ and $b_i = \|\delta_i\|^2 - (w_{i+1} - w_1)$. Then*

$$\pi(\hat{q}, \hat{S}) = \|t\|^2 + w_1 - w_q - u^T G^{-1} b,$$

and, since $\det(G) > 0$,

$$\text{sign}(\pi(\hat{q}, \hat{S})) = \text{sign}\left(\det(G) (\|t\|^2 + w_1 - w_q) - u^T \text{adj}(G) b\right),$$

whose argument is a polynomial of total degree $2k$ in the input coordinates and weights.

Proof. Write $p_S = p_1 + \sum_i \lambda_i \delta_i$; as in the existence proposition, the orthogonality conditions give $G\lambda = b/2$. Then $\|q - p_S\|^2 = \|t\|^2 - 2u^T\lambda + \lambda^T G\lambda$ and $w_S = \|p_S - p_1\|^2 - w_1 = \lambda^T G\lambda - w_1$, so the quadratic terms cancel in the power distance:

$$\pi(\hat{q}, \hat{S}) = \|q - p_S\|^2 - w_q - w_S = \|t\|^2 + w_1 - w_q - 2u^T\lambda = \|t\|^2 + w_1 - w_q - u^T G^{-1}b.$$

Multiplying by $\det(G) > 0$ (positive definite by affine independence) and using $\text{adj}(G) = \det(G) G^{-1}$ gives the claim. $\square$

For $k = 2$ the Gram matrix is the scalar $\|d\|^2$ with $d = b - a$, its adjugate is 1, and the cleared form collapses to a degree-4 polynomial built from three inner products:

$$E = \|d\|^2 (\|t\|^2 + w_a - w_q) - (t \cdot d) (\|d\|^2 + w_a - w_b), \quad \text{sign}(E) = \text{sign}(\pi(\hat{q}, \hat{S})).$$

Remark 14 (Comparison of the two routes). The two formulations trade degree against structure, and the crossover matters for robust implementations. For some combinations of n and k one may be more efficient than the other.

- The cleared form has total degree $2k$ but no artificial construction and no orientation factor at all. For $k = 2$ it is dramatically smaller (for the exact stage of the staged implementation: a buffer bound of 1897 doubles versus 33701 for each of the three basis branches of the D form, and roughly $4\times$ faster in both the filtered and the exact paths); for $k = 3$ the advantage is marginal, and for $k = n + 1$ the D form wins (degree $2k = 8$ versus 5).

Finally, the proposition below on $\text{sign}(w_S - \alpha)$ is this same denominator-clearing technique applied to the radius test. No lower-degree analogue exists: $\det(G)$ is (generically) the exact denominator of w_S , so degree $2k$ is intrinsic. Its exact stage is by far the most expensive one, improvements there must come from the evaluation strategy, not from the algebra.

Proposition 15. *Let $\hat{p}_1, \dots, \hat{p}_k$ be k weighted points in $\mathbb{R}^n$ with $2 \leq k \leq n + 1$, whose centers are affinely independent, and consider their orthogonal hypersphere $\hat{S}$. Define the vector $b \in \mathbb{R}^{k-1}$ with components $b_i = \|p_{i+1} - p_1\|^2 - (w_{i+1} - w_1)$, and the $(k-1) \times (k-1)$ matrix G with components $G_{ij} = (p_{i+1} - p_1) \cdot (p_{j+1} - p_1)$. Then,*

$$\text{sign}(w_S - \alpha) = -\text{sign} \det \begin{pmatrix} 4(w_1 + \alpha) & b^T \\ b & G \end{pmatrix}$$

Proof. By definition of $\hat{S}$, its center p_S must lie in the affine hull of $p_1, \dots, p_k$, so $p_S = p_1 + \sum_{i=2}^k \lambda_i (p_i - p_1)$. Imposing the orthogonality condition $\pi(\hat{S}, \hat{p}_j) = 0 \ \forall j = 1, \dots, k$, we obtain the linear system $G\lambda = \frac{b}{2}$ for the coefficients λ_i . Note that G is the Gram matrix of $(p_i - p_1)$, and by the hypothesis of affine independence, it is positive definite (so $\det(G) > 0$ and is invertible). Also, note that imposing $\pi(\hat{S}, \hat{p}_1) = 0$ implies $w_S + w_1 = \|p_S - p_1\|^2 = \lambda^T G\lambda = b^T G^{-1}b/4$.

On another hand, consider the matrix $\begin{pmatrix} 4(w_1 + \alpha) & b^T \\ b & G \end{pmatrix}$. By the Schur complement formula, we have that its determinant equals $\det(G) \cdot \det(4(w_1 + \alpha) - b^T G^{-1}b)$. By the previous result, this equals $\det(G) \cdot (4w_1 + 4\alpha - 4w_S - 4w_1) = -4(w_S - \alpha)\det(G)$. Since $4\det(G) > 0$, this proves the claim. $\square$

B Linear Programming tests for Voronoi cells

B.1 The LP feasibility predicate

Two applications below — clipping a Voronoi cell against a triangle (in the 2-dimensional oblique coordinates of its plane) and against a tetrahedron (directly in ambient $\mathbb{R}^3$) — reduce to the same question at different ambient dimension n : given n half-space constraints whose boundaries meet at a unique point, on which side of an $(n + 1)$ -th query constraint does that point lie? We isolate this predicate once, at general n , and specialize it to $n = 2$ and $n = 3$ in the sections that follow.

Definition 16 (Constraint). A *constraint* $\mathcal{C}$ in $\mathbb{R}^n$ is a pair $(a, c) \in (\mathbb{R}^n \setminus \{0\}) \times \mathbb{R}$, defining the half-space $\{x \in \mathbb{R}^n \mid a \cdot x \leq c\}$. A point x **strictly satisfies** $\mathcal{C}$ if $a \cdot x < c$, **violates** it if $a \cdot x > c$, and lies on its **boundary** if $a \cdot x = c$.

Definition 17 (Slope determinant). Given n constraints $\mathcal{C}_1, \dots, \mathcal{C}_n$ in $\mathbb{R}^n$ with normals $a_1, \dots, a_n$, their **slope determinant** is

$$\Delta(\mathcal{C}_1, \dots, \mathcal{C}_n) = \det(a_1, \dots, a_n).$$

Proposition 18. $\Delta(\mathcal{C}_1, \dots, \mathcal{C}_n) \neq 0$ if and only if the n boundary hyperplanes of $\mathcal{C}_1, \dots, \mathcal{C}_n$ meet at a unique point p , given by $p = M^{-1}c$, where M has rows $a_1, \dots, a_n$ and $c = (c_1, \dots, c_n)$.

Proof. The boundaries $a_i \cdot x = c_i$, $i = 1, \dots, n$, form the linear system $Mx = c$. This has a unique solution if and only if M is invertible, i.e. $\det M = \Delta \neq 0$, in which case $p = M^{-1}c$. $\square$

Theorem 19 (LP feasibility predicate). Let $\mathcal{C}_1, \dots, \mathcal{C}_n$ be constraints in $\mathbb{R}^n$ with $\Delta(\mathcal{C}_1, \dots, \mathcal{C}_n) \neq 0$, so their boundaries meet at the unique point p of Proposition 18. Let $\mathcal{C}_{n+1} = (a_{n+1}, c_{n+1})$ be a query constraint. Define

$$\Gamma(\mathcal{C}_1, \dots, \mathcal{C}_{n+1}) = \det \begin{pmatrix} a_1 & c_1 \\ \vdots & \vdots \\ a_{n+1} & c_{n+1} \end{pmatrix}, \quad \text{feasible}(\mathcal{C}_1, \dots, \mathcal{C}_n; \mathcal{C}_{n+1}) = \text{sign}(\Delta) \cdot \text{sign}(\Gamma).$$

Then:

- $\text{feasible} > 0 \iff p$ strictly satisfies $\mathcal{C}_{n+1}$.
- $\text{feasible} = 0 \iff p$ lies on the boundary of $\mathcal{C}_{n+1}$ (or $\Delta = 0$, by convention).
- $\text{feasible} < 0 \iff p$ violates $\mathcal{C}_{n+1}$.

Moreover feasible is invariant under any permutation of $\mathcal{C}_1, \dots, \mathcal{C}_n$: transposing two of them flips the sign of both Δ and Γ , leaving their product unchanged.

Proof. Let M have rows $a_1, \dots, a_n$, $c = (c_1, \dots, c_n)$, so $p = M^{-1}c$. Let $V = a_{n+1} \cdot p - c_{n+1}$ be the signed violation of $\mathcal{C}_{n+1}$ at p : $V < 0$ is strict satisfaction, $V > 0$ is violation. Write Γ as the bordered block determinant

$$\Gamma = \det \begin{pmatrix} M & c \\ a_{n+1} & c_{n+1} \end{pmatrix}.$$

Since M is invertible, the Schur complement of M gives

$$\Gamma = \det(M)(c_{n+1} - a_{n+1}M^{-1}c) = \Delta \cdot (c_{n+1} - a_{n+1} \cdot p) = -\Delta \cdot V.$$

Hence $\text{sign}(V) = -\text{sign}(\Delta)\text{sign}(\Gamma)$, and $\text{feasible} := -\text{sign}(V) = \text{sign}(\Delta)\text{sign}(\Gamma)$ has the stated meaning. When $\Delta = 0$ the block factorization is invalid — exactly the degenerate case where the n boundaries fail to meet at a unique point — and we set $\text{feasible} = 0$ by convention. $\square$

Lemma 20 (Reference-point reduction). *Let $A \in \mathbb{R}^n$ be any point. Replacing the offset column $(c_1, \dots, c_{n+1})$ of Γ by $(c_1 - a_1 \cdot A, \dots, c_{n+1} - a_{n+1} \cdot A)$ leaves Γ , and hence feasible , unchanged.*

Proof. The new offset column equals the old one minus $\sum_{j=1}^n A_j(\text{column } j)$, a linear combination of the normal-columns already present in Γ ; adding a multiple of one column to another leaves a determinant unchanged. $\square$

The reference point A carries no geometric meaning of its own — by the lemma above, feasible does not depend on it. Its only role is to lower the polynomial degree of the offset entries: choosing A on the boundary of one of the constraints zeroes that row's offset entirely.

The other recurring object in both applications is the perpendicular bisector between two seeds, which defines a constraint directly in $\mathbb{R}^n$ without reference to any embedding:

Definition 21 (Bisector constraint). Let $s, t \in \mathbb{R}^n$ with $s \neq t$. The **bisector constraint** $\mathcal{S} = (s \rightarrow t)$ is the half-space favoring s in the perpendicular bisector of s, t :

$$\mathcal{S} = (a_{\mathcal{S}}, c_{\mathcal{S}}), \quad a_{\mathcal{S}} = t - s, \quad c_{\mathcal{S}} = \frac{1}{2}(\|t\|^2 - \|s\|^2),$$

i.e. x satisfies $\mathcal{S}$ if and only if $\|x - s\|^2 \leq \|x - t\|^2$.

Lemma 22 (Bisector value). *For $\mathcal{S} = (s \rightarrow t)$ and any $x \in \mathbb{R}^n$,*

$$a_{\mathcal{S}} \cdot x - c_{\mathcal{S}} = \frac{1}{2}(\|x - s\|^2 - \|x - t\|^2).$$

Proof. Expand $\|x - s\|^2 - \|x - t\|^2 = -2s \cdot x + \|s\|^2 + 2t \cdot x - \|t\|^2 = 2(t - s) \cdot x - (\|t\|^2 - \|s\|^2) = 2(a_{\mathcal{S}} \cdot x - c_{\mathcal{S}})$. $\square$

B.2 Cell-triangle intersection

This section specializes Theorem 19 to $n = 2$: clipping a Voronoi cell against a triangle, cast as a feasibility problem in the oblique coordinates of the triangle's plane.

Oblique coordinates and the restriction lemma

Let $\Pi \subset \mathbb{R}^3$ be a plane through $p_0 \in \mathbb{R}^3$, spanned by linearly independent vectors u, v . Every point of Π is $x(\alpha, \beta) = p_0 + \alpha u + \beta v$ for a unique $(\alpha, \beta) \in \mathbb{R}^2$: a one-to-one map $\mathbb{R}^2 \rightarrow \Pi$.

Lemma 23 (Restriction of a half-space to a plane). *Let $\mathcal{H} = (\nu, d)$ be a constraint in $\mathbb{R}^3$ (Definition 16) with $\nu \cdot u, \nu \cdot v$ not both zero. Then $\mathcal{H} \cap \Pi$ is, in the oblique coordinates (α, β) , the planar constraint $\mathcal{C} = (a, c)$ with*

$$a = (\nu \cdot u, \nu \cdot v), \quad c = d - \nu \cdot p_0.$$

Proof. $x(\alpha, \beta)$ satisfies $\mathcal{H}$ iff $\nu \cdot (p_0 + \alpha u + \beta v) \leq d$, i.e. $\alpha(\nu \cdot u) + \beta(\nu \cdot v) \leq d - \nu \cdot p_0$, which is exactly $a \cdot (\alpha, \beta) \leq c$. The hypothesis ensures $a \neq 0$ (ν is not orthogonal to Π), so $\mathcal{C}$ is well-defined. $\square$

Corollary 24 (Planar bisector constraint). *Let $s_1, s_2 \in \mathbb{R}^3$, $s_1 \neq s_2$, with $s_2 - s_1$ not orthogonal to Π . The restriction to Π of the bisector constraint (Definition 21) ($s_1 \rightarrow s_2$) is, in oblique coordinates, $\mathcal{S} = (a, c)$ with*

$$a = ((s_2 - s_1) \cdot u, (s_2 - s_1) \cdot v), \quad c = \frac{1}{2}(\|s_2 - p_0\|^2 - \|s_1 - p_0\|^2).$$

Proof. Apply Lemma 23 with $\nu = s_2 - s_1$, $d = \frac{1}{2}(\|s_2\|^2 - \|s_1\|^2)$ (Definition 21). By Lemma 22 at $x = p_0$, $c_{\mathcal{S}} - \nu \cdot p_0 = \frac{1}{2}(\|p_0 - s_1\|^2 - \|p_0 - s_2\|^2)$, so $c = d - \nu \cdot p_0 = \frac{1}{2}(\|s_2 - p_0\|^2 - \|s_1 - p_0\|^2)$. $\square$

Triangle and seed constraints

Let $\mathbf{T}$ be a triangle with affinely independent vertices $A, B, C \in \mathbb{R}^3$, spanning $\Pi_{\mathbf{T}}$ with oblique coordinates $p_0 = A$, $u = B - A$, $v = C - A$, so $\mathbf{T} = \{x(\alpha, \beta) \mid \alpha \geq 0, \beta \geq 0, \alpha + \beta \leq 1\}$. Equivalently $\mathbf{T}$ is the intersection of three **triangle constraints**, written conventionally as triples (a, b, c) rather than as a vector-offset pair:

$$\mathcal{C}_{\mathbf{T}_0} = (-1, 0, 0), \quad \mathcal{C}_{\mathbf{T}_1} = (0, -1, 0), \quad \mathcal{C}_{\mathbf{T}_2} = (1, 1, 1).$$

For two seeds s_1, s_2 with $s_2 - s_1$ not orthogonal to $\Pi_{\mathbf{T}}$, we call the restriction of $(s_1 \rightarrow s_2)$ to $\Pi_{\mathbf{T}}$ (Corollary 24, $p_0 = A$) a **seed constraint** $\mathcal{S} = (a, b, c)$:

$$a = (s_2 - s_1) \cdot (B - A), \quad b = (s_2 - s_1) \cdot (C - A), \quad c = \frac{1}{2}(\|s_2 - A\|^2 - \|s_1 - A\|^2).$$

Every vertex of $\mathbf{T} \cap \text{cell}(s_1)$ is the intersection of two of these constraints, tested by Theorem 19 at $n = 2$ against a third.

Corollary 25. *For any constraint $\mathcal{C} = (a, b, c)$ and the triangle constraints above, writing D_{ij} for the slope determinant (Definition 17) of constraints i, j :*

$$D_{\mathbf{T}_0, \mathcal{C}} = -b, \quad D_{\mathbf{T}_1, \mathcal{C}} = a, \quad D_{\mathbf{T}_2, \mathcal{C}} = b - a.$$

Proof. Direct evaluation of the 2×2 slope determinant with a triangle constraint's normal as one row, e.g. $D_{\mathbf{T}_0, \mathcal{C}} = \det \begin{pmatrix} -1 & 0 \\ a & b \end{pmatrix} = -b$, and likewise for the other two. $\square$

Proposition 26 (Bound direction). *Let $\mathcal{C}_i, \mathcal{C}_j$ be planar constraints with $D_{ij} = a_i b_j - a_j b_i \neq 0$, so by Proposition 18 ($n = 2$) their boundaries meet at a unique point p_{ij} . Parametrize $\mathcal{C}_i$'s boundary line by $t \mapsto (\alpha_0 + b_i t, \beta_0 - a_i t)$, direction $(b_i, -a_i)$ orthogonal to (a_i, b_i) . Then $\mathcal{C}_j$ restricts this line from below if $D_{ij} > 0$ (a **lower bound**, cutting off $t \rightarrow -\infty$) and from above if $D_{ij} < 0$ (an **upper bound**, cutting off $t \rightarrow +\infty$).*

Proof. Substituting the parametrization into $a_j \alpha + b_j \beta \leq c_j$ gives $-t \cdot D_{ij} \leq c_j - a_j \alpha_0 - b_j \beta_0$. If $D_{ij} < 0$ this reads $t \leq t^*$ ($\mathcal{C}_j$ an upper bound); if $D_{ij} > 0$ it reads $t \geq t^*$ (a lower bound). $\square$

Feasibility across constraint types

We call constraints of the form $\mathcal{S}_i = (a_i, b_i, c_i)$ (SEED) and the fixed $\mathcal{C}_{\mathbf{T}_0}, \mathcal{C}_{\mathbf{T}_1}, \mathcal{C}_{\mathbf{T}_2}$ (TRI) the two families in play. Every vertex of $\mathbf{T} \cap \text{cell}(s)$ is defined by two constraints tested against a third; by Theorem 19, $\text{feasible}(\mathcal{C}_i, \mathcal{C}_j; \mathcal{C}_k) = \text{feasible}(\mathcal{C}_j, \mathcal{C}_i; \mathcal{C}_k)$, so only the unordered definer pair and the query matter. The table below covers every combination that can arise.

Proposition 27. *With $\mathcal{C}_{\mathbf{T}_0} = (-1, 0, 0)$, $\mathcal{C}_{\mathbf{T}_1} = (0, -1, 0)$, $\mathcal{C}_{\mathbf{T}_2} = (1, 1, 1)$, a seed constraint $\mathcal{S} = (a, b, c)$, and two seed constraints $\mathcal{S}_i = (a_i, b_i, c_i)$, $\mathcal{S}_j = (a_j, b_j, c_j)$ with $c_i^* = 2c_i$, $c_j^* = 2c_j$:*

Triple	feasible
$(\mathcal{C}_{\mathbf{T}_i}, \mathcal{C}_{\mathbf{T}_j}; \mathcal{C}_{\mathbf{T}_k})$	$(1 - \delta_{ij})(1 - \delta_{ik})(1 - \delta_{jk})$
$(\mathcal{C}_{\mathbf{T}_0}, \mathcal{C}_{\mathbf{T}_1}; \mathcal{S})$	$\text{sign}(c) = \text{sign}(\ s_2 - A\ ^2 - \ s_1 - A\ ^2)$
$(\mathcal{C}_{\mathbf{T}_1}, \mathcal{C}_{\mathbf{T}_2}; \mathcal{S})$	$\text{sign}(c - a) = \text{sign}(\ s_2 - B\ ^2 - \ s_1 - B\ ^2)$
$(\mathcal{C}_{\mathbf{T}_0}, \mathcal{C}_{\mathbf{T}_2}; \mathcal{S})$	$\text{sign}(c - b) = \text{sign}(\ s_2 - C\ ^2 - \ s_1 - C\ ^2)$
$(\mathcal{C}_{\mathbf{T}_0}, \mathcal{S}; \mathcal{C}_{\mathbf{T}_1})$	$\text{sign}(b) \cdot \text{sign}(c) = -D_{\mathbf{T}_0, \mathcal{S}} \cdot \text{feasible}(\mathcal{C}_{\mathbf{T}_0}, \mathcal{C}_{\mathbf{T}_1}; \mathcal{S})$
$(\mathcal{C}_{\mathbf{T}_1}, \mathcal{S}; \mathcal{C}_{\mathbf{T}_0})$	$\text{sign}(a) \cdot \text{sign}(c) = D_{\mathbf{T}_1, \mathcal{S}} \cdot \text{feasible}(\mathcal{C}_{\mathbf{T}_0}, \mathcal{C}_{\mathbf{T}_1}; \mathcal{S})$
$(\mathcal{C}_{\mathbf{T}_0}, \mathcal{S}; \mathcal{C}_{\mathbf{T}_2})$	$-\text{sign}(b) \cdot \text{sign}(c - b) = D_{\mathbf{T}_0, \mathcal{S}} \cdot \text{feasible}(\mathcal{C}_{\mathbf{T}_0}, \mathcal{C}_{\mathbf{T}_2}; \mathcal{S})$
$(\mathcal{C}_{\mathbf{T}_2}, \mathcal{S}; \mathcal{C}_{\mathbf{T}_0})$	$-\text{sign}(b - a) \cdot \text{sign}(c - b) = -D_{\mathbf{T}_2, \mathcal{S}} \cdot \text{feasible}(\mathcal{C}_{\mathbf{T}_0}, \mathcal{C}_{\mathbf{T}_2}; \mathcal{S})$
$(\mathcal{C}_{\mathbf{T}_1}, \mathcal{S}; \mathcal{C}_{\mathbf{T}_2})$	$-\text{sign}(a) \cdot \text{sign}(c - a) = -D_{\mathbf{T}_1, \mathcal{S}} \cdot \text{feasible}(\mathcal{C}_{\mathbf{T}_1}, \mathcal{C}_{\mathbf{T}_2}; \mathcal{S})$
$(\mathcal{C}_{\mathbf{T}_2}, \mathcal{S}; \mathcal{C}_{\mathbf{T}_1})$	$\text{sign}(b - a) \cdot \text{sign}(c - a) = D_{\mathbf{T}_2, \mathcal{S}} \cdot \text{feasible}(\mathcal{C}_{\mathbf{T}_1}, \mathcal{C}_{\mathbf{T}_2}; \mathcal{S})$
$(\mathcal{C}_{\mathbf{T}_0}, \mathcal{S}_i; \mathcal{S}_j)$	$\text{sign}(b_i) \cdot \text{sign}(b_i c_j^* - c_i^* b_j) = -D_{\mathbf{T}_0, \mathcal{S}_i} \cdot \text{sign}(b_i c_j^* - c_i^* b_j)$
$(\mathcal{C}_{\mathbf{T}_1}, \mathcal{S}_i; \mathcal{S}_j)$	$\text{sign}(a_i) \cdot \text{sign}(a_i c_j^* - c_i^* a_j) = D_{\mathbf{T}_1, \mathcal{S}_i} \cdot \text{sign}(a_i c_j^* - c_i^* a_j)$
$(\mathcal{C}_{\mathbf{T}_2}, \mathcal{S}_i; \mathcal{S}_j)$	$\text{sign}(b_i - a_i) \cdot \text{sign}(b_i c_j^* - c_i^* b_j - a_i c_j^* + c_i^* a_j + 2a_i b_j - 2b_i a_j)$
$(\mathcal{S}_i, \mathcal{S}_j; \mathcal{C}_{\mathbf{T}_0})$	$\text{sign}(a_i b_j - a_j b_i) \cdot \text{sign}(b_j c_i^* - c_i^* b_i)$
$(\mathcal{S}_i, \mathcal{S}_j; \mathcal{C}_{\mathbf{T}_1})$	$\text{sign}(a_i b_j - a_j b_i) \cdot \text{sign}(a_i c_j^* - c_i^* a_j)$
$(\mathcal{S}_i, \mathcal{S}_j; \mathcal{C}_{\mathbf{T}_2})$	$\text{sign}(a_i b_j - a_j b_i) \cdot \text{sign}(b_i c_j^* - c_i^* b_j + c_i^* a_j - a_i c_j^* + 2a_i b_j - 2b_i a_j)$

Proof.

(TRI, TRI; TRI):

Each pair of distinct triangle constraints intersects at a vertex of $\mathbf{T}$, which satisfies the third; repeated constraints yield $D_{ij} = 0$ and (or) make the 3×3 determinant = 0.

(TRI, TRI; SEED):

Vertex A (from $\mathcal{C}_{\mathbf{T}_0}, \mathcal{C}_{\mathbf{T}_1}$): The constraint pair determinant is $D = (-1)(-1) - (0)(0) = 1 > 0$. The 3×3 determinant evaluates to c , giving feasible = $\text{sign}(1) \cdot \text{sign}(c)$.

Vertex B (from $\mathcal{C}_{\mathbf{T}_1}, \mathcal{C}_{\mathbf{T}_2}$): We have $D = (0)(1) - (1)(-1) = 1 > 0$ and the 3×3 determinant equals $c - a$, giving feasible = $\text{sign}(1) \cdot \text{sign}(c - a) = \text{sign}(c - a)$. Expanding the distance difference with A as reference: $\|s_2 - B\|^2 - \|s_1 - B\|^2 = \|s_2 - A\|^2 - \|s_1 - A\|^2 + 2(s_1 - s_2) \cdot (B - A) = 2(c - a)$.

Vertex C (from $\mathcal{C}_{\mathbf{T}_0}, \mathcal{C}_{\mathbf{T}_2}$): We have $D = (-1)(1) - (1)(0) = -1 < 0$ and the 3×3 determinant equals $b - c$, giving feasible = $\text{sign}(-1) \cdot \text{sign}(b - c) = \text{sign}(c - b)$. Similarly: $\|s_2 - C\|^2 - \|s_1 - C\|^2 = \|s_2 - A\|^2 - \|s_1 - A\|^2 + 2(s_1 - s_2) \cdot (C - A) = 2(c - b)$.

(TRI, SEED; TRI):

$(\mathcal{C}_{T_0}, \mathcal{S}; \mathcal{C}_{T_1})$: We have $D_{T_0, \mathcal{S}} = -b$. The 3×3 determinant is: $\begin{vmatrix} -1 & 0 & 0 \\ a & b & c \\ 0 & -1 & 0 \end{vmatrix} = -c$, giving feasible = $\text{sign}(-b) \cdot \text{sign}(-c) = \text{sign}(b) \cdot \text{sign}(c)$.

$(\mathcal{C}_{T_1}, \mathcal{S}; \mathcal{C}_{T_0})$: We have $D_{T_1, \mathcal{S}} = a$. The 3×3 determinant is: $\begin{vmatrix} 0 & -1 & 0 \\ a & b & c \\ -1 & 0 & 0 \end{vmatrix} = c$, giving feasible = $\text{sign}(a) \cdot \text{sign}(c)$.

$(\mathcal{C}_{T_0}, \mathcal{S}; \mathcal{C}_{T_2})$: We have $D_{T_0, \mathcal{S}} = -b$. The 3×3 determinant is: $\begin{vmatrix} -1 & 0 & 0 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix} = c - b$, giving feasible = $\text{sign}(-b) \cdot \text{sign}(c - b)$.

$(\mathcal{C}_{T_2}, \mathcal{S}; \mathcal{C}_{T_0})$: We have $D_{T_2, \mathcal{S}} = b - a$. The 3×3 determinant is: $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ -1 & 0 & 0 \end{vmatrix} = b - c$, giving feasible = $\text{sign}(b - a) \cdot \text{sign}(b - c) = -\text{sign}(b - a) \cdot \text{sign}(c - b)$.

$(\mathcal{C}_{T_1}, \mathcal{S}; \mathcal{C}_{T_2})$: We have $D_{T_1, \mathcal{S}} = a$. The 3×3 determinant is: $\begin{vmatrix} 0 & -1 & 0 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix} = a - c$, giving feasible = $\text{sign}(a) \cdot \text{sign}(a - c) = -\text{sign}(a) \cdot \text{sign}(c - a)$.

$(\mathcal{C}_{T_2}, \mathcal{S}; \mathcal{C}_{T_1})$: We have $D_{T_2, \mathcal{S}} = b - a$. The 3×3 determinant is: $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ 0 & -1 & 0 \end{vmatrix} = c - a$, giving feasible = $\text{sign}(b - a) \cdot \text{sign}(c - a)$.

(TRI, SEED; SEED):

$\mathcal{C}_{T_0}$: We have $D_{T_0, i} = -b_i$. The 3×3 determinant is: $\begin{vmatrix} -1 & 0 & 0 \\ a_i & b_i & c_i \\ a_j & b_j & c_j \end{vmatrix} = b_j c_i - c_j b_i$, giving feasible = $\text{sign}(-b_i) \cdot \text{sign}(b_j c_i - c_j b_i) = \text{sign}(b_i) \cdot \text{sign}(b_i c_j - c_i b_j)$. Substituting $c = c^*/2$ and multiplying by 2, feasible = $\text{sign}(b_i) \cdot \text{sign}(b_i c_j^* - c_i^* b_j)$.

$\mathcal{C}_{T_1}$: We have $D_{T_1, i} = a_i$. The 3×3 determinant is: $\begin{vmatrix} 0 & -1 & 0 \\ a_i & b_i & c_i \\ a_j & b_j & c_j \end{vmatrix} = a_i c_j - c_i a_j$, giving feasible = $\text{sign}(a_i) \cdot \text{sign}(a_i c_j - c_i a_j)$. Substituting $c = c^*/2$, feasible = $\text{sign}(a_i) \cdot \text{sign}(a_i c_j^* - c_i^* a_j)$.

$\mathcal{C}_{T_2}$: We have $D_{T_2, i} = b_i - a_i$. The 3×3 determinant is: $\begin{vmatrix} 1 & 1 & 1 \\ a_i & b_i & c_i \\ a_j & b_j & c_j \end{vmatrix} = (b_i c_j - c_i b_j) - (a_i c_j - c_i a_j) + (a_i b_j - b_i a_j)$, giving feasible = $\text{sign}(b_i - a_i) \cdot \text{sign}(b_i c_j - c_i b_j - a_i c_j + c_i a_j + a_i b_j - b_i a_j)$. Substituting $c = c^*/2$ and multiplying by 2, feasible = $\text{sign}(b_i - a_i) \cdot \text{sign}(b_i c_j^* - c_i^* b_j - a_i c_j^* + c_i^* a_j + 2a_i b_j - 2b_i a_j)$.

(SEED, SEED; TRI):

In all three cases we have $D_{i, j} = a_i b_j - a_j b_i$.

$\mathcal{C}_{T_0}$: The 3×3 determinant is: $\begin{vmatrix} a_i & b_i & c_i \\ a_j & b_j & c_j \\ -1 & 0 & 0 \end{vmatrix} = -(b_i c_j - c_i b_j)$. Substituting $c = c^*/2$: feasible = $\text{sign}(a_i b_j - a_j b_i) \cdot \text{sign}(b_j c_i^* - c_j^* b_i)$.

$\mathcal{C}_{T_1}$: The 3×3 determinant is: $\begin{vmatrix} a_i & b_i & c_i \\ a_j & b_j & c_j \\ 0 & -1 & 0 \end{vmatrix} = a_i c_j - c_i a_j$. Substituting $c = c^*/2$: feasible = $\text{sign}(a_i b_j - a_j b_i) \cdot \text{sign}(a_i c_j^* - c_i^* a_j)$.

$$a_j b_i) \cdot \text{sign}(a_i c_j^* - c_i^* a_j).$$

$\mathcal{C}_{T_2}$: The 3×3 determinant is: $\begin{vmatrix} a_i & b_i & c_i \\ a_j & b_j & c_j \\ 1 & 1 & 1 \end{vmatrix} = (b_i c_j - c_i b_j) - (a_i c_j - c_i a_j) + (a_i b_j - b_i a_j)$, giving:

feasible = $\text{sign}(a_i b_j - a_j b_i) \cdot \text{sign}(b_i c_j - c_i b_j + c_i a_j - a_i c_j + a_i b_j - b_i a_j)$. Substituting $c = c^*/2$ we have: $\text{sign}(a_i b_j - a_j b_i) \cdot \text{sign}(b_i c_j^* - c_i^* b_j + c_i^* a_j - a_i c_j^* + 2a_i b_j - 2b_i a_j)$. $\square$

Robust Seidel's algorithm (summary)

Given a triangle $\mathbf{T}$ and the $n - 1$ seed constraints separating a seed s from its Voronoi neighbors, Seidel's randomized incremental algorithm decides whether $\mathbf{T} \cap \text{cell}(s)$ is non-empty in $O(n)$ expected time, using only D (Definition 17) and feasible (Theorem 19 at $n = 2$) — no coordinate pair (α, β) is ever materialized.

- **Pre-check.** If any pair of triangle constraints has $D = 0$, two edges are parallel and $\mathbf{T}$ is degenerate.
- **Outer loop.** Track a candidate vertex as a pair of active constraints (A, B) , initialized to $(\mathcal{C}_{T_0}, \mathcal{C}_{T_1})$. Process the seed constraints in random order, skipping $\mathcal{C}_i$ whenever $\text{feasible}(A, B; \mathcal{C}_i) \geq 0$.
- **Inner loop.** Otherwise the new vertex must lie on $\mathcal{C}_i$'s boundary line: solve a 1D LP there, represented as an interval (lo, hi) of constraint identifiers. Proposition 26 assigns each previously-processed constraint to the lo or hi slot by the sign of D , tightened by comparing against the current occupant via feasible; the interval is empty iff $\text{feasible}(\mathcal{C}_i, \text{lo}; \text{hi}) < 0$. If so, the triangle and cell do not intersect. Otherwise set $(A, B) = (\mathcal{C}_i, \text{lo})$ and continue.
- **Degenerate sub-cases.** A triangle edge parallel to $\mathcal{C}_i$'s line ($D = 0$) either permits the whole line or forbids it outright, decided by one feasible call against the known finite endpoint. Three triangle boundaries concurrent at a point already known to violate $\mathcal{C}_i$ reject immediately. Both require no machinery beyond feasible itself.

B.3 Cell-tetrahedron intersection

This section specializes Theorem 19 to $n = 3$, directly in ambient $\mathbb{R}^3$ — no chart or restriction step is needed here, unlike §B.2. It answers the same question as §B.2, $\text{cell}(s) \cap X$ for X now a tetrahedron, but to a different depth: §B.2 only needs the intersection's *existence*, so Seidel's algorithm stops at a boolean. Here, downstream mesh reconstruction needs the intersection's actual *shape*, so every vertex of $\text{cell}(s) \cap \mathbf{tet}$ is classified instead — i.e. the cell is clipped against the tetrahedron.

T and S constraints

Let a tetrahedron have vertices P, Q, R, W , and write its face PQR (with W the opposite vertex) as a **T constraint**

$$\mathcal{T} = (n_T, d_T), \quad n_T = (Q - P) \times (R - P), \quad d_T = n_T \cdot P,$$

with the sign of (n_T, d_T) flipped if necessary so that W satisfies $\mathcal{T}$ (the tetrahedron interior is the feasible side); n_T has degree 2 and d_T degree 3. Between the cell's own seed s and a neighboring seed t , the **S constraint** is exactly the bisector constraint ($s \rightarrow t$) of Definition 21, taken directly in $\mathbb{R}^3$ rather than restricted to a plane:

$$a_S = t - s \quad (\text{degree 1}), \quad c_S = \frac{1}{2}(\|t\|^2 - \|s\|^2) \quad (\text{degree 2}).$$

Every vertex of $\text{cell}(s) \cap \mathbf{tet}$ is the intersection of three T/S constraints, tested by Theorem 19 at $n = 3$ against a fourth.

Reference point and a zero-free Γ

By Lemma 20, feasible does not depend on the reference point A ; we use that freedom purely to reduce polynomial degree. **Standing convention:** take A to be a vertex of a T-face among the three vertex-definers (a vertex on the shared edge, when two T-faces are definers). This zeroes that row's offset in Γ outright, and turns every S-row offset into $c = \|t - A\|^2 - \|s - A\|^2$ — dropping the positive factor $\frac{1}{2}$ carried by c_S , which affects no sign.

Remark 28 (Zero-free implementation). A row zeroed by this convention should never be built as a literal 0 entry: a floating-point filter cannot bound a constant-zero leaf. Cofactor-expanding Γ along the offset column instead drops every zero-offset row automatically, leaving a sum of (nonzero offset) $\times$ (3×3 minor of normals) terms. All formulas below are already stated in this zero-free form.

TTS crossings

Two T-faces meet along a tetrahedron edge with endpoints A, B ; an S constraint ($s \rightarrow t$) may cross it. Write $g_X = \|X - s\|^2 - \|X - t\|^2$ (twice the bisector value of Lemma 22, hence affine in X), $h_X = \|X - s\|^2 - \|X - u\|^2$ for a second S constraint ($s \rightarrow u$), and $w_X = n_c \cdot (X - P_c)$ for a query T-face with vertex P_c and interior-oriented normal n_c .

Lemma 29 (Existence). *Parametrizing the edge as $x(\lambda) = A + \lambda(B - A)$, the bisector crosses the segment interior iff $\text{sign}(g_A) \neq \text{sign}(g_B)$, at $\lambda^* = g_A / (g_A - g_B) \in (0, 1)$: g affine gives $g(x(\lambda)) = g_A + \lambda(g_B - g_A)$, root $\lambda^* = g_A / (g_A - g_B)$, placed strictly in $(0, 1)$ exactly when g_A, g_B have opposite signs.*

Lemma 30 (Own tetrahedron is automatic). *If the crossing exists, it automatically satisfies every face of the tetrahedron the edge AB belongs to: A, B are vertices of a convex tetrahedron, so the segment AB , and hence $x(\lambda^*)$, lies inside it.*

Lemma 31 (Ordering two crossings). *For two S constraints $(s \rightarrow t_1), (s \rightarrow t_2)$ crossing the same edge at λ_1, λ_2 (with $g_{iX} = \|X - s\|^2 - \|X - t_i\|^2$),*

$$\text{sign}(\lambda_1 - \lambda_2) = \text{sign}(g_{1B}g_{2A} - g_{1A}g_{2B}) \cdot \text{sign}(g_{1A} - g_{1B}) \cdot \text{sign}(g_{2A} - g_{2B}),$$

substituting $\lambda_i = g_{iA} / (g_{iA} - g_{iB})$ into $\lambda_1 - \lambda_2$ and clearing denominators. Orders two bisectors crossing the same tetrahedron edge, e.g. inside the 1D sub-problem of an edge clip.

Slope determinants for TSS and SSS

Two S constraints $(s \rightarrow t_1), (s \rightarrow t_2)$ meeting a T-face T_1 (normal n_{T_1}) have slope determinant $\Delta_{TSS} = \det[t_1 - s; t_2 - s; n_{T_1}]$, degree 4. Three S constraints $(s \rightarrow t_1), (s \rightarrow t_2), (s \rightarrow t_3)$ meet at the circumcenter p of the Delaunay tetrahedron (s, t_1, t_2, t_3) , with

$$\Delta_{SSS} = \det[t_1 - s; t_2 - s; t_3 - s] = \text{orient}_3(t_1, t_2, t_3, s),$$

immediate from the definition of orient_3 (Theorem 4) with s as the query point; degree 3. Both are nonzero for a genuine tetrahedron. Take the reference point A on T_1 (resp. on the query T-face) and write $c_i = \|t_i - A\|^2 - \|s - A\|^2$.

Proposition 32 (Feasibility for a tetrahedron clip). *Let V be a tetrahedron vertex and T' a query T-face, possibly different from a vertex-defining one:*

Triple	feasible	deg
$(T, T, T; \mathcal{S})$	$-\text{sign}(\ V - s\ ^2 - \ V - t\ ^2)$	2
$(T, T, T; T')$	1 if T' is the face opposite V , else 0	—
$(T, T, \mathcal{S}; \mathcal{S}')$	$-\text{sign}(g_A h_B - g_B h_A) \cdot \text{sign}(g_A - g_B)$	4
$(T, T, \mathcal{S}; T')$	$-\text{sign}(g_A w_B - g_B w_A) \cdot \text{sign}(g_A - g_B)$	5
$(T, \mathcal{S}, \mathcal{S}; \mathcal{S}')$	$\text{sign}(\Delta_{TSS}) \cdot \text{sign}(\Gamma_{TSS, \mathcal{S}})$	4/6
$(T, \mathcal{S}, \mathcal{S}; T')$	$\text{sign}(\Delta_{TSS}) \cdot \text{sign}(\Gamma_{TSS, T})$	4/7
$(\mathcal{S}, \mathcal{S}, \mathcal{S}; \mathcal{S}')$	$-\text{sign}(\text{test_insphere}(s, t_1, t_2, t_3; u))$	4
$(\mathcal{S}, \mathcal{S}, \mathcal{S}; T)$	$\text{sign}(\Delta_{SSS}) \cdot \text{sign}(\Gamma_{SSS, T})$	3/6

Where two degrees are shown they are $\deg \Delta / \deg \Gamma$. Only *test_insphere* and the orient_3 -combinatorics of the first block are reused from Appendix A; the rest are new, reducing to Theorem 19 and Lemma 22, which also underlie §B.2.

Proof.

(a) TTT — a tetrahedron vertex

A tetrahedron vertex V trivially lies on the three faces meeting at it and strictly inside the opposite face, so against a T query the answer is combinatorial. Against an S constraint $(s \rightarrow t)$, Lemma 22 at $x = V$ gives $\text{feasible} = -\text{sign}(a_S \cdot V - c_S) = -\text{sign}(\frac{1}{2}(\|V - s\|^2 - \|V - t\|^2))$.

(b) TTS — a bisector crosses a tetrahedron edge

Against a second S constraint $(s \rightarrow u)$: h is affine, so substituting λ^* (Lemma 29) and clearing the denominator, $h(x(\lambda^*)) = (g_A h_B - g_B h_A) / (g_A - g_B)$, giving $\text{feasible} = -\text{sign}(h(x(\lambda^*)))$. Against an arbitrary query T-face: the signed position of the crossing relative to it is $n_c \cdot x(\lambda^*) - d_c = w_A + \lambda^*(w_B - w_A)$, since $n_c \cdot A - d_c = w_A$ and $n_c \cdot (B - A) = w_B - w_A$; substituting λ^* as before gives the stated 2×2 -determinant form. When the query face belongs to the crossing's own tetrahedron this reduces to the combinatorial fact of Lemma 30 — no computation is needed there.

(c) TSS — a Voronoi edge meets a tetrahedron face

Apply Theorem 19 to $(\mathcal{S}_1, \mathcal{S}_2, T_1)$. Against $\mathcal{S} = (s \rightarrow u)$ with $c_u = \|u - A\|^2 - \|s - A\|^2$: the reference-point convention zeroes the n_{T_1} -row offset, and cofactor-expanding Γ along the offset column (Remark 28) gives

$$\Gamma_{TSS,S} = -c_1 \det[t_2 - s; n_{T_1}; u - s] + c_2 \det[t_1 - s; n_{T_1}; u - s] + c_u \Delta_{TSS},$$

the last minor being Δ_{TSS} itself: degree 6. Against a second, unrelated face T_2 (vertex P_2 , normal n_{T_2} , $w = n_{T_2} \cdot (P_2 - A)$), A lying on T_1 only leaves the offset column $(\frac{1}{2}c_1, \frac{1}{2}c_2, 0, w)$ — a genuine mixture of scales, since the query offset $d_{T_2} - n_{T_2} \cdot A = w$ carries no compensating $\frac{1}{2}$. Doubling the whole column (sign-preserving) clears it to $(c_1, c_2, 0, 2w)$; cofactor-expanding along it gives

$$\Gamma_{TSS,T} = -c_1 \det[t_2 - s; n_{T_1}; n_{T_2}] + c_2 \det[t_1 - s; n_{T_1}; n_{T_2}] + 2w \Delta_{TSS},$$

degree 7. When A can also be chosen on T_2 (e.g. T_1, T_2 share a tetrahedron edge and A is taken on it), $w = 0$ and the last term drops.

(d) SSS — a Voronoi vertex (circumcenter)

Against a fourth S constraint $(s \rightarrow u)$: feasible = $-\text{sign}(a_S \cdot p - c_S) = \text{sign}(\|p - u\|^2 - \|p - s\|^2)$ by Lemma 22. Since p is the circumcenter, $\|p - s\| = \|p - t_i\| = R$; writing $\hat{S} = (p, R^2)$ for the (unweighted) circumsphere, $\pi(\hat{u}, \hat{S}) = \|p - u\|^2 - R^2$ has the same sign, and by §A.2, $\text{sign}(\pi(\hat{u}, \hat{S})) = -\text{test_insphere}(s, t_1, t_2, t_3; u)$ (output 1 there means u lies *inside* the sphere, the opposite of satisfying $\mathcal{S}$). Against a T-face with vertices A, Q, R , normal n_T , reference point A , and $c_i = \|t_i - A\|^2 - \|s - A\|^2$: applying Theorem 19 to the three S constraints and query T, the T-row offset vanishes and each S-row offset becomes c_i , so cofactor-expanding Γ along the offset column (Remark 28) drops the T term and leaves

$$\Gamma_{SSS,T} = -c_1 \det[t_2 - s; t_3 - s; n_T] + c_2 \det[t_1 - s; t_3 - s; n_T] - c_3 \det[t_1 - s; t_2 - s; n_T],$$

degree 6. □

Conventions

- *Face orientation.* A T constraint used as a *query* needs its interior side fixed: multiply by $\sigma = \text{sign}(\text{orient}_3(P_1, P_2, P_3, W))$, with W the opposite tetrahedron vertex and (P_1, P_2, P_3) the query face's vertices in the exact order used to build its normal $n = (P_2 - P_1) \times (P_3 - P_1)$ — a permutation of the face vertices flips n and hence σ .
- *Reference point.* A is a face vertex (an edge vertex when two T constraints are definers or share a common face); by Lemma 20 the predicate value does not depend on it, so no cross-face coordination of A is required.
- *Degeneracy.* feasible = 0 means a vertex lies exactly on the query's boundary. Resolve with a single global rule (symbolic perturbation or a lexicographic order) keyed on constraint identities — seed and tetrahedron-vertex indices, never coordinates — so that every pair of cells or clips sharing that vertex classifies it identically.

B.4 Surface reconstruction: edge ordering

Clipping produces, per cell, the vertices of every cell $\cap$ **tet** piece. Reconstructing the cell's boundary as a watertight mesh requires stitching the surviving faces into closed loops; every

step is combinatorial (keyed on constraint identities) except one metric decision: **ordering the vertices that lie on a shared facet edge**. Two facets P, Q meet along a line $\ell = P \cap Q$; every vertex on ℓ must be placed in the same order by both facets, or a crack opens between them. This reduces entirely to Theorem 19.

Setup

A surface vertex is, as before, a triple of constraint boundaries $\{C_1, C_2, C_3\}$. A facet is a single constraint P ; the vertices on P are those triples containing it. Two facets P, Q share the line $\ell = P \cap Q$ with direction $d = n_P \times n_Q \neq 0$. A vertex on ℓ has P, Q among its three constraints, distinguished by its third: write $u = \{P, Q, R_u\}$, $v = \{P, Q, R_v\}$. Ordering along ℓ is the sign of $d \cdot (u - v)$.

Theorem 33 (Edge ordering reduces to feasibility). *Let $u = \{P, Q, R_u\}$, $v = \{P, Q, R_v\}$ lie on $\ell = P \cap Q$, with slope determinants $\Delta_u, \Delta_v \neq 0$, and $d = n_P \times n_Q$. Then*

$$\text{ord}(u, v) := \text{sign}(d \cdot (u - v)) = -\text{sign}(\Delta_v) \cdot \text{feasible}(P, Q, R_u; R_v).$$

ord is antisymmetric in u, v , hence a consistent total order along ℓ ; the two facets sharing ℓ evaluate the identical Δ 's and Γ 's, so they subdivide it identically.

Proof. u, v both satisfy the boundaries of P, Q , so $u - v$ is parallel to ℓ : $u - v = \mu d$ for a scalar μ , and $\text{sign}(d \cdot (u - v)) = \text{sign}(\mu)$. Dotting with n_{R_v} and using $n_{R_v} \cdot v = c_{R_v}$,

$$n_{R_v} \cdot u - c_{R_v} = \mu (n_{R_v} \cdot d) = \mu \Delta_v, \quad \Delta_v = \det[n_P; n_Q; n_{R_v}].$$

The left side is the signed violation V of R_v at u . By Theorem 19 applied to the triple (P, Q, R_u) with query R_v , $\Gamma_{u;R_v} = -\Delta_u V$, so $\text{sign}(V) = -\text{sign}(\Delta_u) \text{sign}(\Gamma_{u;R_v}) = -\text{feasible}(P, Q, R_u; R_v)$, giving $\text{sign}(\mu) = -\text{sign}(\Delta_v) \text{feasible}(P, Q, R_u; R_v)$. Swapping u, v swaps two rows of Γ (flipping its sign) and exchanges $\Delta_u \leftrightarrow \Delta_v$, so $\text{ord}(v, u) = -\text{ord}(u, v)$. $\square$

Since Δ_v is v 's own slope determinant — already computed to classify v 's type in §B.3 — every ordering costs exactly one feasible evaluation and one already-known sign.

Catalogue

The line $\ell = P \cap Q$ falls into three kinds by the types of P, Q (one ruled block per kind below). Within a block, u 's type is $\{P, Q, R_u\}$ and v 's type is $\{P, Q, R_v\}$ — read off the same two columns, since P, Q are fixed for the whole block. u 's type together with the query R_v selects the row of Proposition 32 that evaluates $\text{feasible}(P, Q, R_u; R_v)$ in Theorem 33's formula; v 's type supplies $\text{sign}(\Delta_v)$, the other factor, already computed to classify v in §B.3.

line $P \cap Q$	R_u	R_v (query)	u 's type	v 's type	row of Prop. 32
S $\cap$ S (Voronoi edge)	S	S	SSS	SSS	$(\mathcal{S}, \mathcal{S}, \mathcal{S}; \mathcal{S}')$
	S	T	SSS	TSS	$(\mathcal{S}, \mathcal{S}, \mathcal{S}; T)$
	T	S	TSS	SSS	$(T, \mathcal{S}, \mathcal{S}; \mathcal{S}')$
	T	T	TSS	TSS	$(T, \mathcal{S}, \mathcal{S}; T')$
S $\cap$ T (bisector/face)	S	S	TSS	TSS	$(T, \mathcal{S}, \mathcal{S}; \mathcal{S}')$
	S	T	TSS	TTS	$(T, \mathcal{S}, \mathcal{S}; T')$
	T	S	TTS	TSS	$(T, T, \mathcal{S}; \mathcal{S}')$
	T	T	TTS	TTS	$(T, T, \mathcal{S}; T')$
T $\cap$ T (tetrahedron edge)	S	S	TTS	TTS	$(T, T, \mathcal{S}; \mathcal{S}')$
	S	T	TTS	TTT	$(T, T, \mathcal{S}; T')$
	T	S	TTT	TTS	$(T, T, T; \mathcal{S})$
	T	T	TTT	TTT	edge endpoints, combinatorial

Table 1: Every edge ordering reduces to a row of Proposition 32, times the sign of v 's own slope determinant. The only row §B.3 did not already need for the clip itself is $(T, T, \mathcal{S}; T')$ in its general form: a clip vertex's query face always belongs to its own tetrahedron (Lemma 30 suffices there), but here the two facets sharing ℓ generally come from different tetrahedra.

Degeneracy

$\text{ord}(u, v) = 0$ means u, v coincide on ℓ (cospherical Voronoi vertices, or a Voronoi vertex lying exactly on a face). Resolve with the same index-keyed rule as §B.3: break ties by the lexicographically-sorted constraint-index tuple of the two vertices, identically for both facets sharing ℓ , collapsing them to a zero-length edge with no coordinate welding required.

References

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